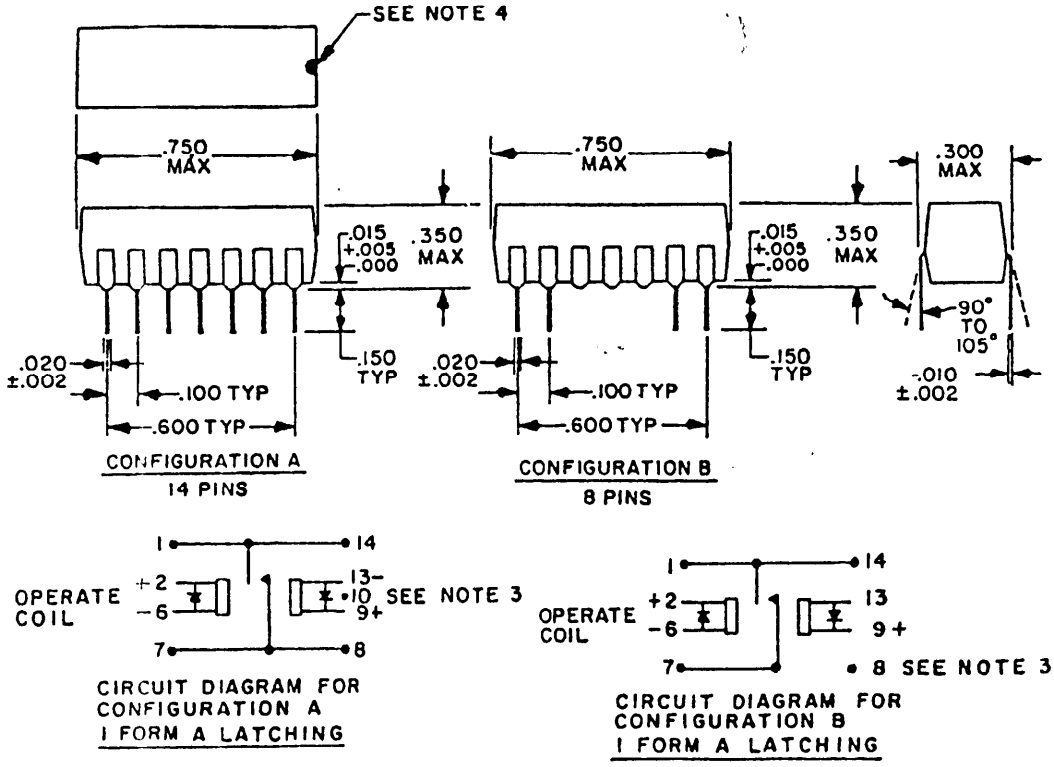
MILITARY SPECIFICATION SHEET

RELAYS, REED, DRY, DUAL IN-LINE PACKAGE (DIP), LATCHING

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-83516.



INCHES	MM
.002	0.05
.005	0.13
.010	0.25
.015	0.38
.020	0.51
.100	2.54
.150	2.81
.300	7.62
.350	8.89
.600	15.24
.750	19.05

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Relays are available with electrostatic shielding at pin 10 for configuration A and at pin 8 for configuration B.
4. A position identifier (shape optional) shall be placed on the top of each relay either centered between pins 1 and 14 or directly over pin 1.

FIGURE 1. Outline drawing.

(D) denotes changes

FSC 5945

Source: https://assist.dla.mil -- Downloaded: 20261O-23T08:31Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers and applicable characteristics.

Dash number						Configuration (see fig. 1)	DC coil data at 25°C								
Without shielding			With shielding				Resistance (minimum) ohms			Operate voltage (maximum) V dc			Power dissipation (nominal) mW		
Coil voltage			Coil voltage				Coil voltage			Coil voltage			Coil voltage		
5	12	24	5	12	24		5	12	24	5	12	24	5	12	24
001	002	003	004	005	006	A	675	900	4,140	3.4	7.5	15.0	70	100	100
007	008	009	010	011	012	B	675	900	4,140	3.4	7.5	15.0	70	100	100

REQUIREMENTS:

DESIGN AND CONSTRUCTION:

Dimensions and configurations: See figure 1.

Diode characteristics: Diodes shall be JANTX as a minimum. 1/

Peak inverse voltage: 75.0 V dc.

CONTACT DATA:

Arrangement: See figure 1.

Load ratings:

Contacts	V dc (max)	mA (max)	VA (max)
Form A	100	500	10

Contact resistance:

Before life: 0.130 ohm maximum.

During and after life: 1.0 ohm maximum.

Group A inspection: 0.200 ohm maximum.

Contact stability: Applicable.

COIL DATA: See table I.

Duty rating: Continuous.

Coil power dissipation: See table I.

Operate time (including bounce): 1 ms nominal.

1/ WARNING: Reverse polarity on coil terminals will destroy diode.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:31Z
Check the source to verify that this is the current version before use.

ELECTRICAL DATA:

Dielectric withstanding voltage: MIL-STD-202, method 301. 2/

Test points:

Between all mutually insulated components: 500 V dc.

Across contacts: 250 V dc.

Insulation resistance: 10,000 MΩ minimum.

Test potential: 100 V dc ±10%.

Test points: Between contacts.

Coil transient suppression: 1.0 V maximum negative transient.

(D) Capacitance, pF maximum:

Across open contacts, no shield:	1.25.
Across open contacts, shield guarded:	0.25.
Contact to coil, no shield:	3.0.
Contact to coil, shield guarded:	2.0.

ENVIRONMENTAL DATA:

Thermal EMF: Not applicable.

Thermal shock: Applicable.

Shock (specified pulse): Applicable.

Vibration, high frequency: Applicable.

Salt spray (corrosion): Applicable.

Moisture resistance: Applicable.

Magnetic interference: Relays are subject to interaction with stray magnetic fields. To keep within operating parameters, relays shall be mounted no closer than .500 inch (12.7 mm) of each other. Greater spacing is required for stronger magnetic fields.

Resistance to soldering heat: Applicable.

PHYSICAL:

Enclosure: Molded plastic.

Operating temperature range: -40°C to +105°C.

Lead integrity: MIL-STD-883, method 2004, test condition B2; the force to be applied shall be 8.0 ±0.5 ounces.

Solderability: Applicable.

Life: Applicable.

(D) Weight: 3.3 grams maximum.

Part number: M83516/2-(dash number from table I).

2/ Connect coil leads together to avoid damage to the diodes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:31Z
Check the source to verify that this is the current version before use.

QUALIFICATION:

Qualification inspection and sample size: See table II.

TABLE II. Qualification inspection and sample size.

M83516/2-	Number of samples
005 or 011	13 units plus 1 open unit. Qualification inspection as applicable.
004 and 006 or 010 and 012	2 units each part number. Qualification inspection group I and life.

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 85

(Project 5945-F553-2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:31Z
Check the source to verify that this is the current version before use.